

Dawson G. Butler

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<https://dawsgbutler-png.github.io/dawsgbutler.github.io/index.html>

PROFESSIONAL SUMMARY

Mechanical Engineering student at the University of Tennessee with hands-on experience in robotic systems assembly, CAD-based design, and electromechanical troubleshooting. Autodesk-certified with a practical background in manufacturing, automation, and technical design seeking to apply engineering fundamentals in a robotics or technology-driven environment.

EDUCATION

University of Tennessee, Tickle College of Engineering

Bachelor of Science in Mechanical Engineering | Aerospace Engineering Minor. Dean's List
2025. GPA: 3.27.

Expected May 2028

EXPERIENCE

Robotics Engineering Technician

2023 – 2026

LUCID Bots

Charlotte, NC

- Assembled ~10 robotic cleaning systems per week through full mechanical build, wiring, testing, and quality inspection.
- Diagnosed and resolved fit issues, wiring faults, assembly errors, and quality control concerns during production.
- Helped clear a backlog of 50 robots toward on-time delivery through system builds and production flow improvements.
- Rebuilt and tuned a 3D printer for prototyping and internal parts production.

College Project Engineer Intern, Virtual Design and Construction

May 2026 – Present

Gaylor Electric

Charlotte, NC

- Execute VDC workflows for large-scale AI data center projects, supporting electrical room builds across configurations of up to 5 colos with 4 cells each plus admin rooms.
- Build and revise Revit model content from project prints and existing plan information.
- Create and refine Revit families and parametric models for modular skid and mounting plate components.
- Identify drawing conflicts, constructability issues, and coordination problems prior to field installation.

PROJECTS

AI Trading Decision System

Python, Alpaca API, Automation

- Built a Python-based system to scan stocks, evaluate trade opportunities, execute paper trades, and track performance.
- Built decision logic using technical indicators, market conditions, and risk management rules.
- Implemented automated reporting, backtesting, and order management.

Construction Safety Analytics Platform

QR Codes, Dashboard Design, Process Improvement

- Designed a QR code-based near-miss reporting system to identify jobsite hazards before recordable incidents occur.
- Developed a dashboard concept for tracking hazard trends, response times, and corrective actions.

ACTIVITIES & LEADERSHIP

Delta Tau Delta Fraternity

2024 to Present

Director of Standards and Construction Project Lead

- Lead house construction and improvement projects; managed an \$8,000 Homecoming budget coordinating materials, labor, and schedules. Designed and fabricated custom event displays, overseeing design, fabrication, and delivery.

CERTIFICATIONS

- Autodesk Certified User: Inventor 2024
- Autodesk Certified User: AutoCAD 2023

SKILLS

Manufacturing: Robotic Assembly, Electromechanical Systems, Wiring & Troubleshooting, 3D Printing, Mechanical Fabrication, Hand & Power Tools

Programming: Python, MATLAB, API Integration, Automation, Backtesting

CAD & Design: Autodesk Revit, Autodesk Inventor, AutoCAD, Parametric Modeling, Revit Family Creation, Mechanical Drafting

Construction & VDC: BIM/VDC Coordination, Bluebeam Revu, Autodesk Construction Cloud (ACC), Shop & Installation Drawings, Plan Review